

Today: Quiz 8

Review

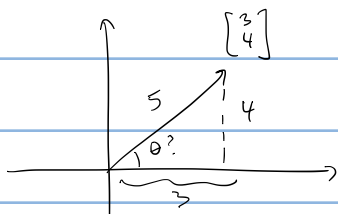
Online evaluations

9.2. 6.7.) $A = \begin{bmatrix} 2 & -1 & -1 \\ 2 & 1 & 1 \\ -1 & 1 & -1 \end{bmatrix}$

$$\begin{bmatrix} \boxed{2} & -1 & -1 & | & 1 & 0 & 0 \\ \textcircled{2} & 1 & 1 & | & 0 & 1 & 0 \\ \textcircled{-1} & 1 & -1 & | & 0 & 0 & 1 \end{bmatrix} \begin{array}{l} R_1 - R_2 \rightarrow R_1 \\ R_1 + 2R_3 \rightarrow R_3 \end{array}$$

$$\begin{bmatrix} \textcircled{-1} & -1 & 1 & | & 1 & 0 & 0 \\ \textcircled{-2} & 2 & 2 & | & 1 & 1 & 0 \\ \textcircled{1} & -3 & 1 & | & 0 & 2 \end{bmatrix} \begin{array}{l} 2R_1 - R_2 \rightarrow R_1 \\ R_2 + 2R_3 \rightarrow R_3 \end{array} \begin{bmatrix} 2 & 0 & \textcircled{0} & | & 1 & 1 & 0 \\ 0 & -2 & \textcircled{2} & | & 1 & -1 & 0 \\ 0 & 0 & \textcircled{-8} & | & 3 & -1 & 4 \end{bmatrix} \xrightarrow{4R_2 + R_3 \rightarrow R_2} \begin{bmatrix} 2 & 0 & 0 & | & 1 & 1 & 0 \\ 0 & -2 & 0 & | & 7 & -5 & 4 \\ 0 & 0 & -8 & | & 3 & -1 & 4 \end{bmatrix}$$

$$\begin{array}{l} \frac{1}{2}R_1 \rightarrow R_1 \\ -\frac{1}{2}R_2 \rightarrow R_2 \\ -\frac{1}{8}R_3 \rightarrow R_3 \end{array} \begin{bmatrix} 1 & 0 & 0 & | & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & | & -\frac{7}{2} & \frac{5}{2} & -2 \\ 0 & 0 & 1 & | & -\frac{3}{8} & \frac{1}{8} & -\frac{1}{2} \end{bmatrix} \quad A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{7}{2} & \frac{5}{2} & -2 \\ -\frac{3}{8} & \frac{1}{8} & -\frac{1}{2} \end{bmatrix}$$



$$\sin^{-1}\left(\frac{4}{5}\right) \approx 0.927$$

$$\approx 53.13^\circ$$

$$\sin \theta = \frac{y}{r} \Rightarrow y = r \sin \theta$$

$$\cos \theta = \frac{x}{r} \Rightarrow x = r \cos \theta$$

11.) $x = 1 \cos 120^\circ$

$$y = 1 \sin 120^\circ$$

$$75. \quad L = \begin{bmatrix} 2 & 4 \\ 0.3 & 0 \end{bmatrix}$$

$$(a) \quad \tau = 2 \quad \Delta = -0.3 \cdot 4 = -1.2$$

$$\lambda^2 - 2\lambda - 1.2 = 0$$

$$\lambda = \frac{2 \pm \sqrt{4 + 1.2 \cdot 1}}{2}$$

$$= 1 \pm \sqrt{5.2}$$

$$\approx 2.48, -0.48$$

↑

$$(c) \quad \begin{bmatrix} 2 - 2.48 & 4 \\ 0.3 & -2.48 \end{bmatrix}$$

$$0.48u_1 = 4u_2$$

$$\vec{u} = \begin{bmatrix} 4 \\ 0.48 \end{bmatrix} \quad \frac{4}{4 + 0.48} \approx 89\% \\ \frac{0.48}{4 + 0.48} \approx 11\%$$

Chapter 5

5.8 Antiderivatives

Chapter 6

6.3 Area, Cumulative Change, Average Values

Chapter 7

7.1 Substitution

7.2 IBP

7.3 Rational Function & Partial Fractions

7.4 Improper Integrals

7.6 Taylor Approximation

Chapter 8

8.1 Solving Differential (separable)

$$\frac{ds}{dt} = f(t)g(s)$$

Additional Topic: 1st-order linear DEs

$$\frac{ds}{dt} + p(t)s = q(t)$$

8.2 Equilibria and Stability

Chapter 9

9.1 Linear Systems

9.2 Matrices

9.3 Linear Maps, Eigenvectors, and Eigenvalues

9.4 Analytic Geometry